

CUTTING SAMPLE & LITHOLOGY



GEOLOGICAL ANALYSIS EQUIPMENT

1. Cutting / cores preparation and description tools

- ☐ Sieves
- ☐ Sample trays
- ☐ Ceramic plates
- ☐ Chisel, hammer
- ☐ Metal Can
- ☐ Sample Bag, unwashed & dry
- ☐ Alumunium Foil
- ☐ Wooden Box
- ☐ Chisel, hammer, needles, tweezers, etc)

2. Standard set of chemical

- ☐ ALIZARIN RED SOLUTION (Analisa kandungan Limestone atau Dolomit)
- ☐ CALCIUM CARBIDE
- ☐ CALCIUM CARBONATE
- ☐ TRICHLOROTHENE (Analisa Hidrokarbon)
- ☐ CHLOROFORM
- ☐ HYDROCHLORIC ACID 10% & 25%
- ☐ PHENOLPHTHALEIN SOLUTION (Mengetahuin adanya kandungan Cement)
- ☐ ACETONE
- ☐ ISO-PROPYL ALCOHOL

3. Geological Equipment

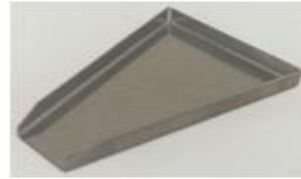
- ☐ Binocular Microscope (10x-40x Variable zoom)
- ☐ Fluoroscope, ultraviolet tubes viewing box
- ☐ Auto calcimetry kit
- ☐ Shale density kit.
- ☐ Samples drying oven and hood.
- ☐ Dip Tank

Cutting Analysis

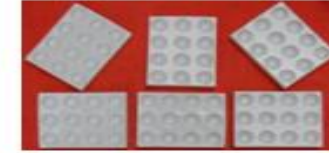
Sample examination items provided in each unit



Unwashed sample & plastic bag



Sample Tray



pestle mortar & Spot dithes



Glass plasma bottle & pipet test



Drys sample & box



Wooden core box



Alumunium foil



Metal Can



Sedimentary comperator



Colour chart

Cutting Analysis



Mikroskop



UV box



Auto Calcimetry



Oven



Shale Density Meter



Dip Tank

CATCHING & PREPARATION OF CUTTING SAMPLE

3. Cutting sampai permukaan di shale shaker



2. Waktu yang di perlukan Cutting sampai di permukaan, Per 30 feet, 10 feet (tergantung program)

Lag Time



1. Drilling Formasi batuan, cutting terangkat bersamaan dengan lumpu pemboran



Cutting Analysis

COLLECT CUTTING SAMPLE

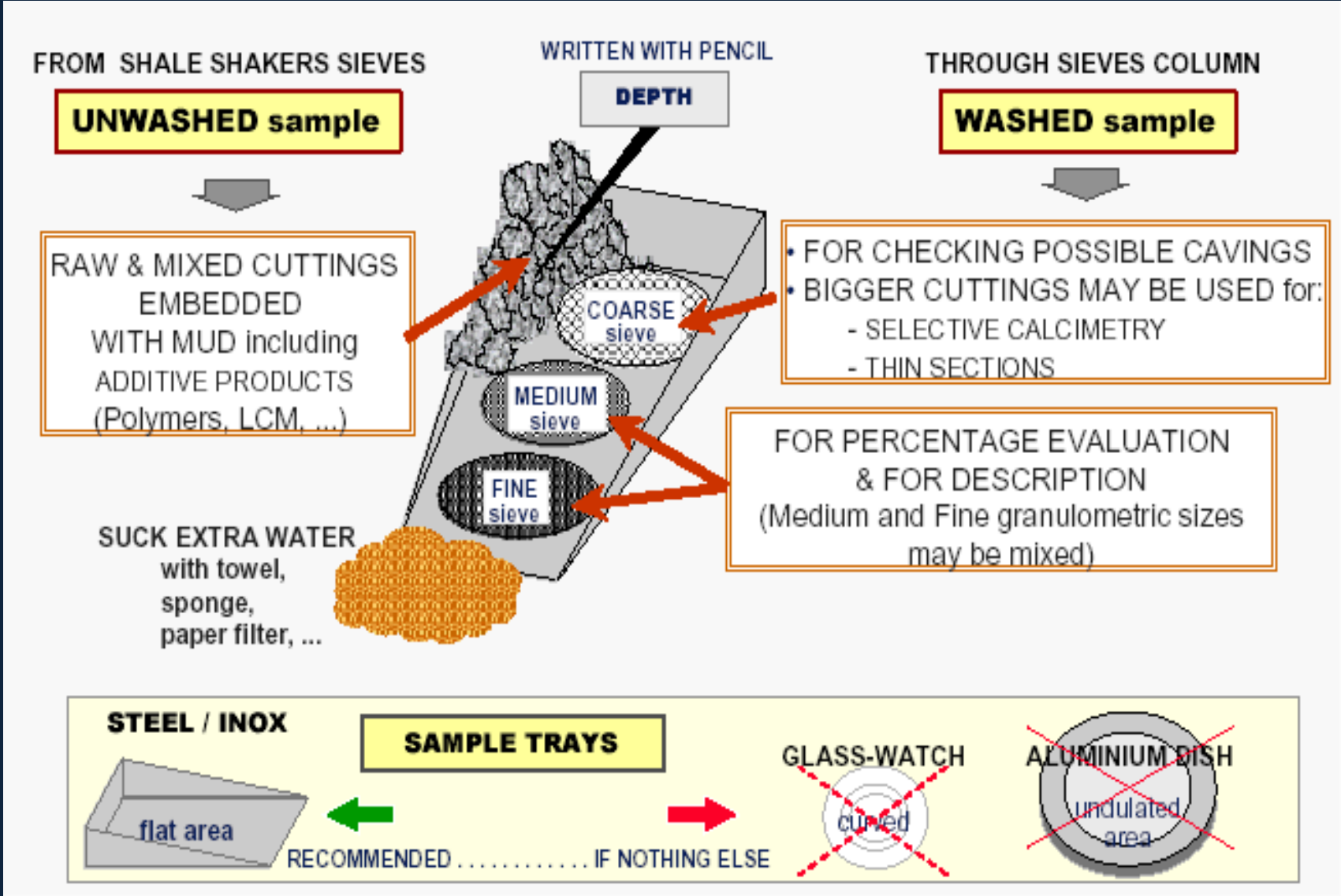


Cutting sampai permukaan di shale shaker

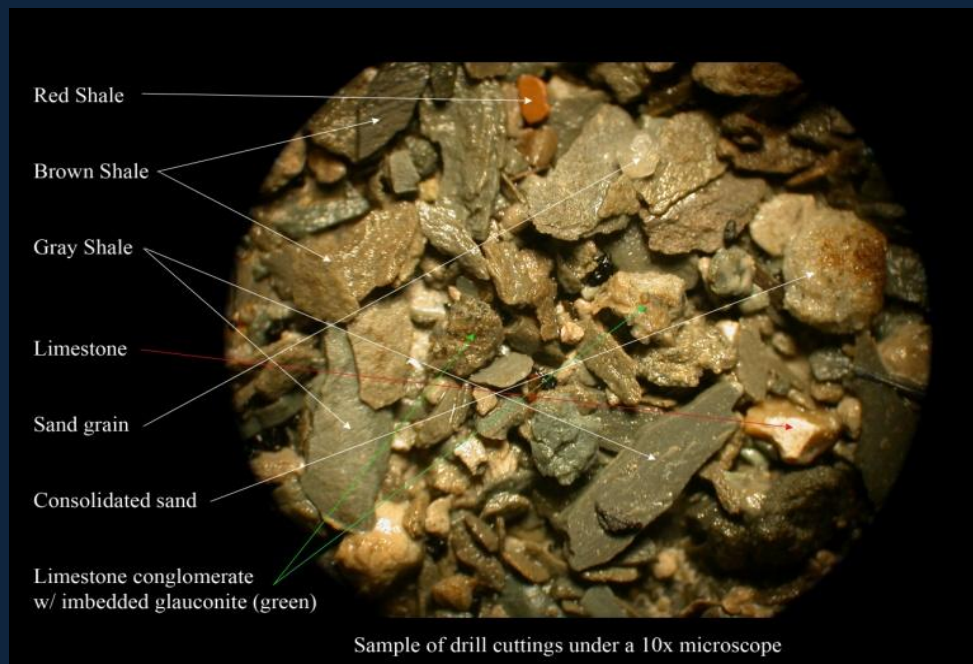
Pencucian cutting untuk di lihat di mikroskop sebagai Washed Sample (Sample yang di Cuci) dan di Dry sample (Sample di keringkan)



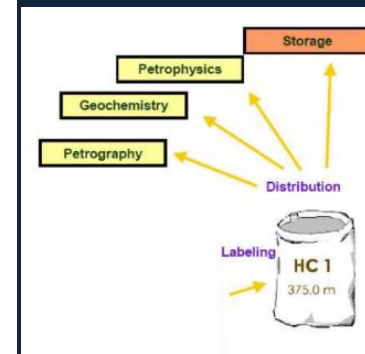
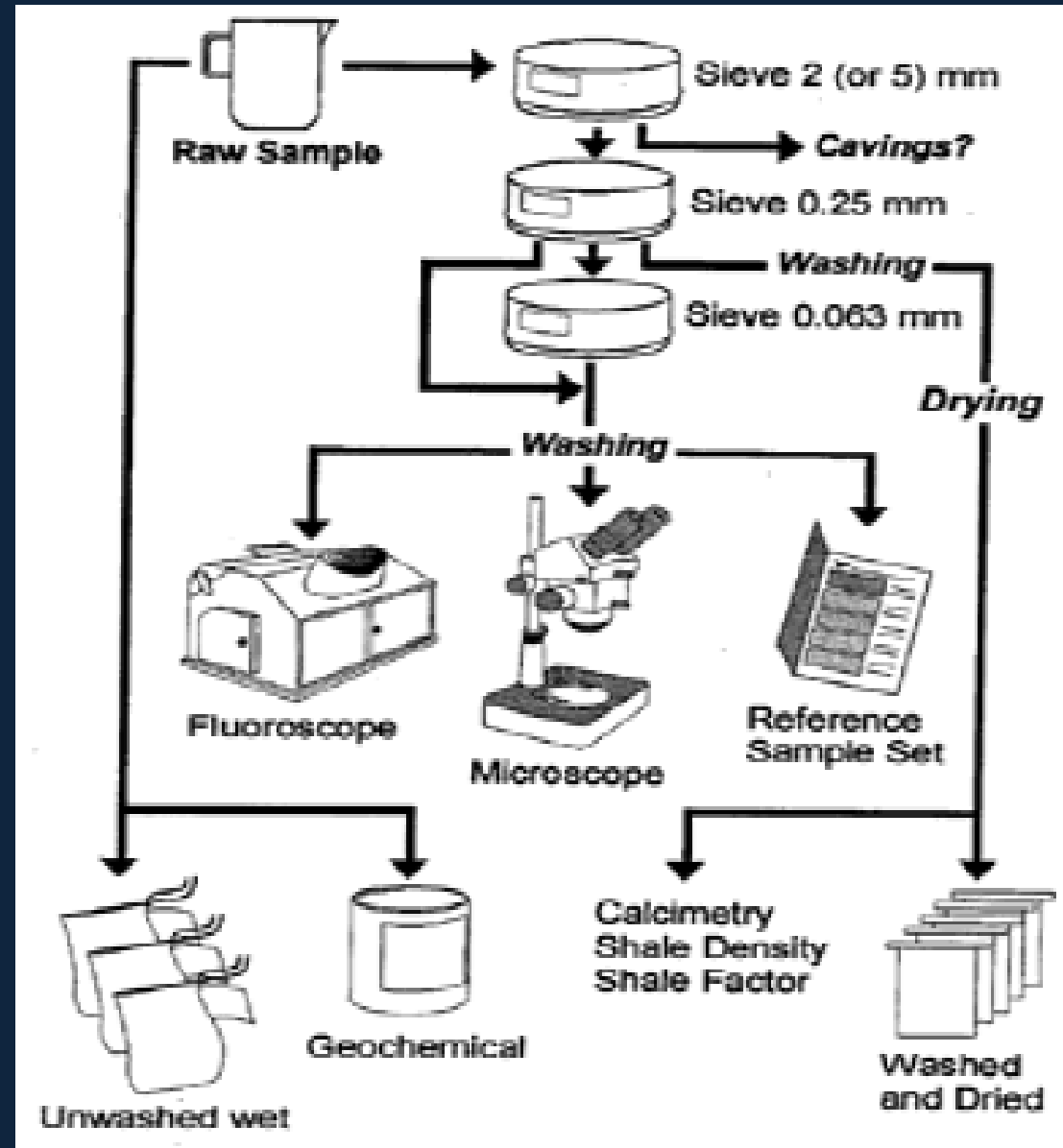
Cutting Analysis



Cutting Analysis



Cutting Analysis



Cutting Analysis

Cuttings vs. Cavings



LOOK FOR



Shale Shaker

- Volume
- Size
- Type



CUTTINGS



CAVINGS

Cutting Analysis



Cutting Analysis

SAMPLE COLLECTION

Pengambilan sample ditentukan Drilling program dari company. Pada umumnya **sample regular** diambil :

- ❑ 26" Hole sample per 10 m. (kira2 per 30 ft)
- ❑ 12-1/4" Hole, sample per 5 m (kira2 per 15 ft)
- ❑ 8-1/2" Hole, sample per 2 m (kira2 per 10 ft)

Pengambilan sample juga tergantung jenis sumur dan kedalaman lapisan yang prospek. Untuk sumur eksplorasi pada trayek lobang 12-1/4" dilakukan pengambilan sample regular per 2 m.

Demikian pula untuk sample Geokimia biasanya diambil per 30 m (tergantung program)

WASHED SAMPLE

Sample yang telah dicuci untuk didiskripsi

Sample dikeringkan dengan microwave/oven dan dibungkus dengan sample envelope (+/- 100-200 gr)

Sample diambil dengan menggunakan saringan 80 mesh, 120 mesh dan 200 mesh

Sample terakhir sebelum cabut pipa harus diambil untuk didiskripsi walaupun bukan sample regular

UNWASH SAMPLE

Sample basah (tanpa dicuci) dimasukkan dalam cloth bag (+/- 300-400 gr) untuk analisa Paleontology. Setiap pengambilan sample, sisa-sisa sample di penampungan harus disemprot bersih. Sehingga sample yang berikutnya benar-benar sample baru.

Cutting Analysis

Batuan dibagi menjadi 3 kelompok utama: batuan beku, batuan sedimen, dan batuan metamorf.

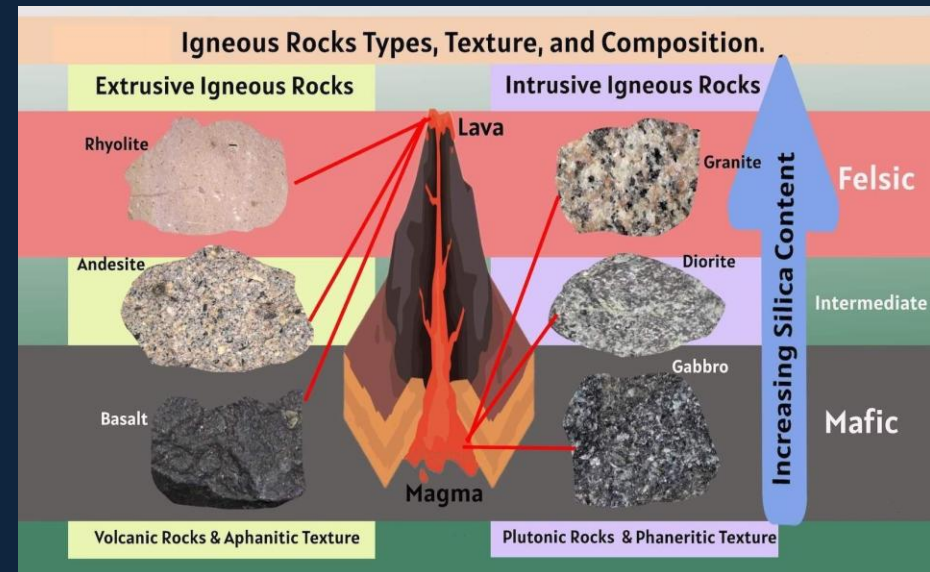
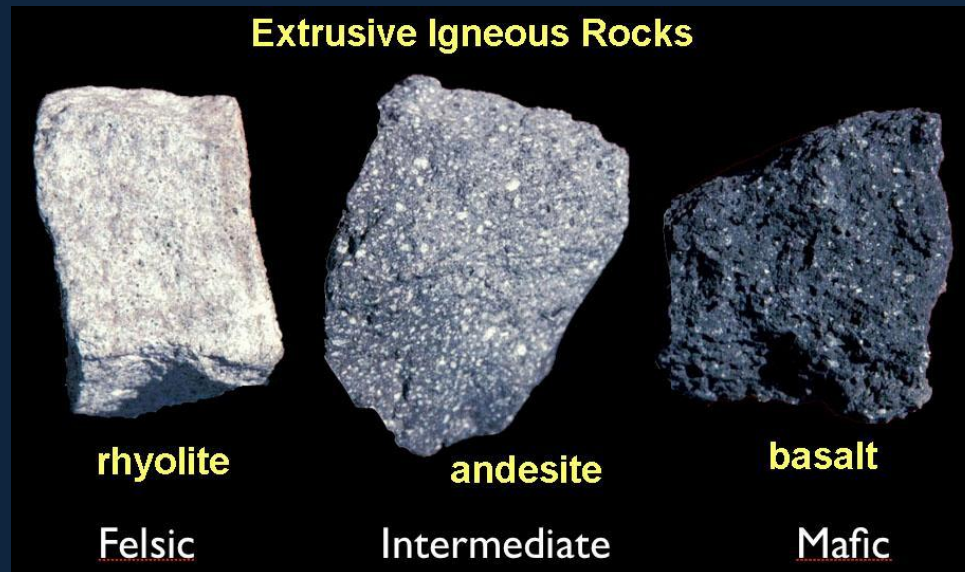
1. Batuan Beku — Igneous Rock

Terbentuk dari pendinginan dan kristalisasi magma atau lava.

A. Beku dalam / Intrusive (Plutonic) — magma membeku perlahan di bawah permukaan sehingga kristalnya relatif besar. Contoh: Granite, Diorite, Gabbro, Granodiorite.

B. Beku luar / Extrusive (Volcanic) — lava membeku cepat di permukaan sehingga ukuran kristalnya halus atau bahkan seperti kaca. Contoh: Rhyolite, Andesite, Basalt, Dacite, Obsidian, Pumice.

C. Hypabyssal / Subvolcanic — membeku pada kedalaman dangkal, berada di antara intrusive dan extrusive. Contoh: Dolerite/Diabase, Microdiorite, Porphyry.



Cutting Analysis

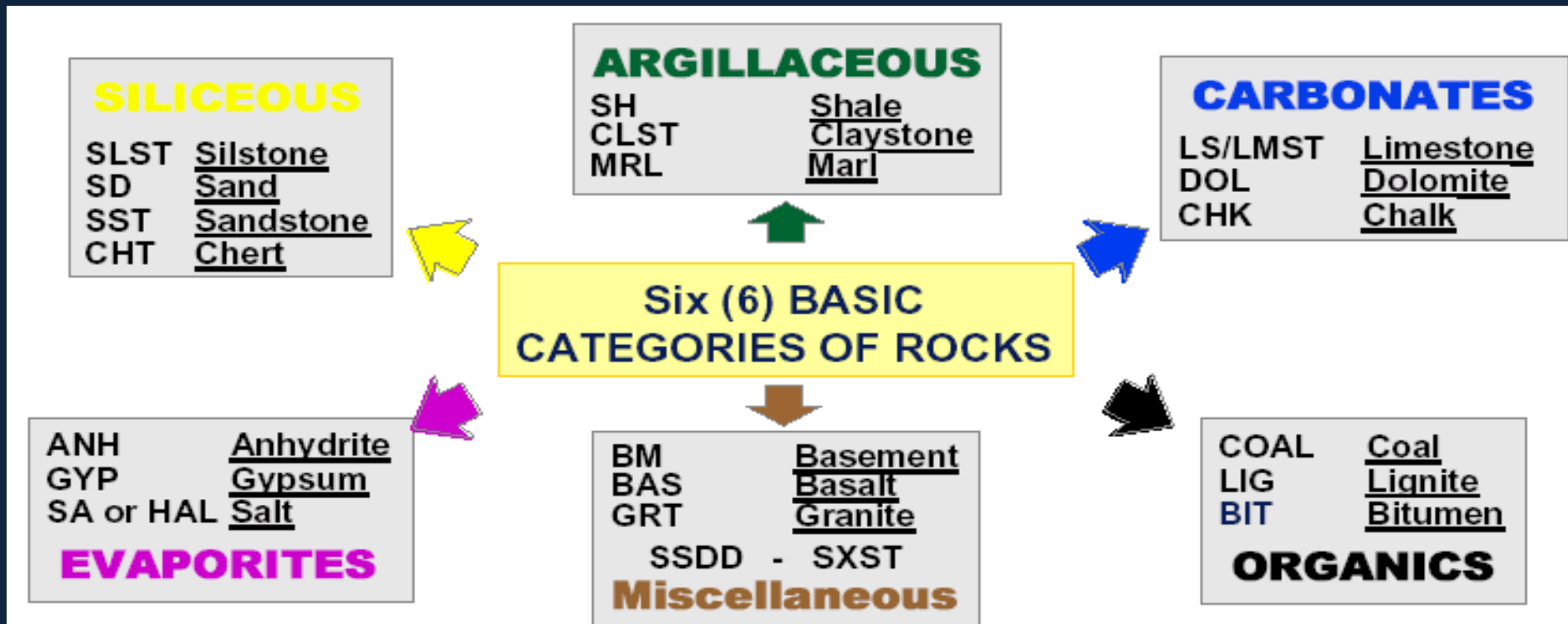
2. Batuan Sedimen — Sedimentary Rock

Terbentuk melalui pelapukan → erosi → transportasi → pengendapan → kompaksi → sementasi/diagenesis, atau melalui presipitasi kimia dan akumulasi material biologis.

A. Klastik / Detrital — berasal dari fragmen batuan sebelumnya. Contoh berdasarkan ukuran butir: Conglomerate, Breccia, Sandstone, Siltstone, Claystone/Shale.

B. Kimia / Chemical — terbentuk karena presipitasi mineral dari larutan. Contoh: Rock Salt, Gypsum, sebagian Limestone, Chert.

C. Biokimia/Organik — berasal dari akumulasi material organisme atau material organik. Contoh: Coal, Fossiliferous Limestone, Chalk, Diatomite.



Cutting Analysis

3. Batuan Metamorf — Metamorphic Rock

Terbentuk ketika batuan asal (protolith) mengalami perubahan akibat **temperatur, tekanan, fluida, dan deformasi**, tetapi tidak sampai melebur sepenuhnya.

A. Foliated Metamorphic Rock — mempunyai orientasi mineral atau perlapisan/foliasi. Urutan metamorfisme yang umum:

Shale → Slate → Phyllite → Schist → Gneiss

Semakin ke kanan, secara umum **metamorphic grade** semakin tinggi.

B. Non-Foliated Metamorphic Rock — tidak menunjukkan foliasi yang jelas. Contoh: **Limestone → Marble**, **Sandstone → Quartzite**, serta **Hornfels** pada metamorfisme kontak.

FOLIATED METAMORPHIC ROCKS



gneiss



schist



phyllite



slate

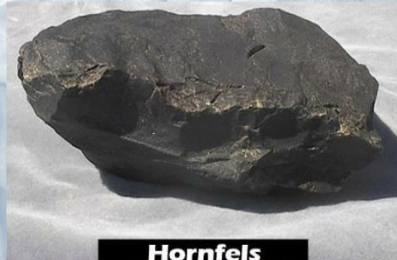
Non-foliated rocks



Marble



Quartzite



Hornfels

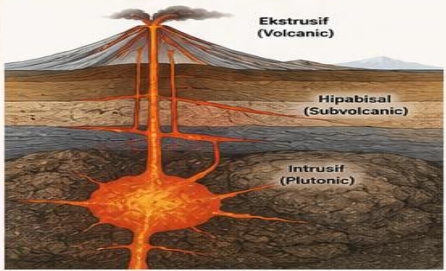
Metamorphic Rock Identification Chart					
TEXTURE	FOLIATION	COMPOSITION	TYPE	PARENT ROCK	ROCK NAME
Foliated	slaty	mica	Regional	Mudstone	Slate
	phyllitic	quartz, mica, chlorite	Regional	Mudstone	Phyllite
	schistose	mica, quartz	Regional	Slate	Schist
	schistose	amphibole, plagioclase	Regional	Basalt or Gabbro	Amphibolite
	gneissic banding	feldspar, mica, quartz	Regional	Schist	Gneiss
Non-Foliated		carbon	Contact or Regional	Bituminous Coal	Anthracite Coal
		quartz, rock fragments	Contact or Regional	Conglomerate	Metaconglomerate
		calcite	Contact or Regional	Limestone	Marble
		quartz	Contact or Regional	Sandstone	Quartzite

KLASIFIKASI BATUAN & LINGKUNGAN PENGENDAPANNYA

BERDASARKAN PROSES PEMBENTUKAN

1. BATUAN BEKU (IGNEOUS ROCK)

Terbentuk dari pembekuan dan kristalisasi magma atau lava. Proses pembekuan dapat terjadi di dalam kerak bumi (intrusif) atau di permukaan (ekstrusif).



A. INTRUSIF / PLUTONIK (MEMBEKU DI DALAM BUMI)				
Nama Batuan	Komposisi	Warna Umum	Tekstur	Contoh Batuan
Granit	Felsik	Terang	Kasar	
Diorit	Intermediat	Sedang	Kasar	
Gabbro	Mafik	Gelap	Kasar	
B. HIPABISAL / SUBVOLKANIK (MEMBEKU DI KEDALAMAN DANGKAL)				
Nama Batuan	Komposisi	Warna Umum	Tekstur	Contoh Batuan
Dolerit / Diabas	Mafik	Gelap	Halus - menengah	
Porfiri Andesit	Intermediat	Abu-abu	Porfiritik	
Mikrodiorit	Intermediat	Sedang	Halus	
C. EKSTRUSIF / VULKANIK (MEMBEKU DI PERMUKAAN)				
Nama Batuan	Komposisi	Warna Umum	Tekstur	Contoh Batuan
Riolit	Felsik	Terang	Halus / Kaca	
Andesit	Intermediat	Abu-abu	Halus - porfiritik	
Basalt	Mafik	Gelap	Halus	
Obsidian	Felsik (kaca)	Hitam	Kaca	
Pumis	Felsik (vesikular)	Terang	Berongga	

LINGKUNGAN TERBENTUK

Lingkungan Vulkanik (Permukaan Bumi)
Terbentuk dari letusan gunungapi. Lava keluar ke permukaan dan membeku cepat.



Lingkungan Intrusif (Dalam Kerak Bumi)
Magma membeku sangat lambat di dalam kerak bumi sehingga kristal berukuran besar.



Lingkungan Hipabisal (Dekat Permukaan)
Magma membeku pada kedalaman relatif dangkal (retakan, gang, sill) dengan kristal umumnya lebih halus.

2. BATUAN SEDIMEN (SEDIMENTARY ROCK)

Terbentuk dari hasil pelapukan batuan yang diangkut, diendapkan, kemudian mengalami kompaksi dan sementasi (litifikasi), atau dari presipitasi kimia dan akumulasi organisme.



A. SEDIMEN KLASTIK / DETRITAL (MEKANIK)				
Nama Batuan	Ukuran Butir	Ciri-ciri	Contoh Batuan	Lingkungan Pengendapan
Konglomerat	>2 mm (membulat)	Fragmen membulat, tersemen kuat		Sungai berenergi tinggi, kipas aluvial
Breksi	>2 mm (bersudut)	Fragmen bersudut tajam		Kipas aluvial, lereng curam, jatuhkan batuan
Batupasir (Sandstone)	0,0625 - 2 mm	Butir pasir, sering berlapis		Pantai, delta, sungai, dune (gunung), laut dangkal
Batulanau (Siltstone)	0,004 - 0,0625 mm	Butir halus seperti tepung		Dataran banjir, danau, laut dangkal tenang
Batulempung (Shale)	<0,004 mm	Sangat halus, berlapis tipis		Laut dalam, danau tenang, laguna, dataran banjir
B. SEDIMEN KIMIA (PRESIPITASI KIMIA)				
Batugamping Kimia (Limestone)	Kristal kalsit	Sering bereaksi dengan HCl		Laut dangkal hangat, platform karbonat
Gypsum	Kristal gipsum	Putih, lunak		Danau kering (evaporit), laguna tertutup
Garam Batu (Rock Salt)	Kristal halit	Putih transparan, berasa asin		Cekungan evaporit, laut tertutup
Kersikan (Chert)	Mikrokristal silika	Sangat keras, pecahan konkoidal		Laut dalam, radiolaria, spongia
C. SEDIMEN ORGANIK / BIOKIMIA				
Batubara (Coal)	Material tumbuhan	Hitam, mudah terbakar		Rawa, hutan bakau, dataran banjir
Batugamping Fosil (Fossiliferous Limestone)	Cangkang organisme	Mengandung fosil		Laut dangkal jernih
Kapur (Chalk)	Foraminifera	Putih, halus		Laut dalam terbuka
Diatomit	Sisa diatom	Ringan, berpori		Danau, laut tenang

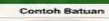
LINGKUNGAN PENGENDAPAN UTAMA



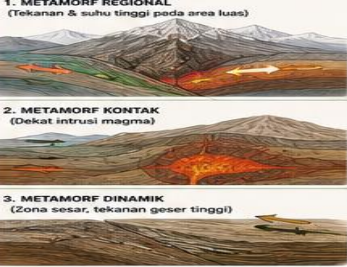
3. BATUAN METAMORF (METAMORPHIC ROCK)

Terbentuk dari perubahan batuan asal karena pengaruh suhu tinggi, tekanan besar, dan/atau fluida aktif, tanpa peleburan total.



A. METAMORF BERFOLIASI (BERLAPIS / BERORIENTASI)					
Tingkat Metamorf	Batuan Asal	Nama Batuan Metamorf	Ciri-ciri	Contoh Batuan	
Rendah	Shale / Lempung	Slate (Batu Sabak)	Balokan slaty, halus		
Menengah Rendah	Shale	Phyllite	Mengkilap, berfoliasi halus		
Menengah Tinggi	Shale	Schist (Sekis)	Kristal terlihat, berfoliasi kuat		
Tinggi	Shale / Granit	Gneiss	Pita terang-gelap, berfoliasi kuat		
 Grade Metamorf Meningkat					
	B. METAMORF NON-FOLIASI (TIDAK BERLAPIS)				
	Batuan Asal	Nama Batuan Metamorf	Ciri-ciri	Contoh Batuan	
	Batugamping (Limestone)	Marble (Pualam)	Kristal kalsit, bereaksi HCl		
	Batupasir Kuarsa (Quartz Sandstone)	Quartzite	Sangat keras, kristal kuarsa		
Batugamping / Shale	Hornfels	Keras, masif, rekristalisasi			
Batubara	Anthracite	Hitam mengkilap, keras			

LINGKUNGAN METAMORFISME



RINGKASAN KLASIFIKASI BATUAN



Cutting Analysis

APA ITU BATUAN SEDIMEN?

Batuan sedimen terbentuk dari hasil pelapukan batuan yang diangkut, diendapkan, lalu mengalami kompaksi dan sementasi (litifikasi). Umumnya berlapis dan sering mengandung fosil.



KLASIFIKASI BATUAN SEDIMEN

SEDIMENTARY ROCK CLASSIFICATION

SIMBOL LITOLOGI (UMUM)



Konglomerat



Breksi



Batupasir



Batulanau



Batulempung



Batugamping

1. BATUAN SEDIMEN KLASTIK (DETTRITAL)

Tersusun dari fragmen/klastik batuan atau mineral hasil pelapukan yang diendapkan secara mekanik.

A. BERDASARKAN UKURAN BUTIR

KONGLOMERAT



- Fragmen membulat (>2 mm)
- Tersemen kuat
- Umum di sungai, kipas aluvial

BREKSI



- Fragmen bersudut (>2 mm)
- Tersemen kuat
- Umum di lereng, runtuhan, aliran debris

BATUPASIR



- Ukuran 0,0625 – 2 mm
- Terasa kasar
- Sering berlapis
- Porositas baik

BATULANAU



- Ukuran 0,004 – 0,0625 mm
- Terasa halus
- Sering berlapis tipis
- Porositas sedang

BATULEMPUNG (SHALE)



- Ukuran <0,004 mm
- Sangat halus, mudah pecah lembaran
- Kedap air
- Potensi shale gas/oil

B. BERDASARKAN KOMPOSISI

BATUPASIR KUARSA



- Terutama kuarsa (>90%)
- Warna cerah
- Kuat dan kompak

BATUPASIR FELDSPATIK



- Banyak feldspar & fragmen batuan
- Warna lebih gelap
- Kurang stabil

BATUPASIR LITIK



- Banyak fragmen batuan
- Warna gelap
- Umum pada lingkungan tektonik aktif

BATUPASIR GLAUKONITIK



- Mengandung mineral glaukonit (hijau)
- Umum di laut dangkal
- Sering fosil

BATU TUF



- Fragmen abu vulkanik
- Lunak, berpori
- Umum di lingkungan vulkanik

C. STRUKTUR SEDIMEN KLASTIK

PERLAPASAN SILANG (CROSS BEDDING)



- Terbentuk oleh arus
- Indikasi arah arus purba

PERLAPASAN SEJAJAR (PLANAR BEDDING)



- Endapan tenang/merata
- Umum di danau/laut

GRADASI BUTIR (GRAIN SIZE GRADING)



- Butir kasar di bawah, halus di atas
- Indikasi arus berkurang

RIPPLE MARK



- Gelombang kecil di dasar sedimen
- Umum di pantai/laut dangkal

FOSIL



- Sisa organisme
- Penentu umur & lingkungan pengendapan

2. BATUAN SEDIMEN KIMIA (NON-KLASTIK)

Terbentuk dari endapan kimiawi melalui presipitasi mineral dari air.

BATUGAMPING (LIMESTONE)



- Komposisi utama kalsit (CaCO_3)
- Bereaksi dengan HCl (berbuih)
- Umum di laut dangkal
- Sering mengandung fosil

DOLOMIT



- Komposisi dolomit $\text{CaMg}(\text{CO}_3)_2$
- Bereaksi lemah dengan HCl
- Lebih keras dari batugamping
- Umum di platform karbonat

BATU GARAM (ROCK SALT)



- Halit (NaCl)
- Rasa asin
- Larut dalam air
- Terbentuk di lingkungan evaporit

GIPSUM



- Komposisi $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- Lebih lunak (gores kuku)
- Umum di lingkungan evaporit
- Dapat larut air

BATUAN SILIKA (CHERT)



- Komposisi silika (SiO_2)
- Keras, pecahan konkoidal
- Umum di laut dalam
- Sering berasosiasi fosil radiolaria

BATU BESI (IRONSTONE)



- Kaya oksida besi ($\text{Fe}_2\text{O}_3/\text{Fe}_3\text{O}_4$)
- Warna merah – coklat gelap
- Terbentuk di lingkungan laut
- Endapan kimia atau diagenetik

3. BATUAN SEDIMEN ORGANIK

Terbentuk dari akumulasi sisa organisme.

BATUBARA



- Dari tumbuhan darat
- Warna hitam
- Mengandung karbon tinggi
- Sumber energi fosil

BATU GAMPING ORGANOGENIK



- Tersusun dari cangkang organisme
- Banyak fosil
- Umum di laut dangkal

DIATOMIT



- Dari cangkang diatom (alga)
- Sangat berpori
- Warna putih – krem
- Digunakan untuk filtrasi

BATU LEMPUNG ORGANIK



- Kaya bahan organik
- Warna gelap
- Potensi sumber hidrokarbon

CONTOH LINGKUNGAN PENGENDAPAN



Sungai & Aluvial



Pantai & Shallow Marine



Laut Dalam (Deep Marine)



Danau & Lagoon



Evaporit (Sabkha)

FAKTOR YANG MEMPENGARUHI PEMBENTUKAN BATUAN SEDIMEN



Sumber Material (Asal Batuan)



Transportasi (Arus, Gelombang, Angin, Es, Gravitasi)



Pengendapan (Deposition)



Kompaksi (Pressure)



Sementasi (Cementation)



Diagenesis & Litifikasi

CARA IDENTIFIKASI LAPANGAN



Warna & kilap



Ukuran butir & tekstur



Kekerasan (uji kuku/pisau)



Reaksi dengan HCl (untuk karbonat)



Struktur sedimen



Kandungan fosil



Berat jenis



Lingkungan pengendapan

KETERANGAN UKURAN BUTIR (SKALA WENTWORTH)

Klasifikasi	Ukuran (mm)	Contoh Batuan
Bongkah	>256	Konglomerat, Breksi
Kerikil	2 – 256	Konglomerat, Breksi
Pasir	0,0625 – 2	Batupasir
Lanau (Silt)	0,004 – 0,0625	Batulanau
Lempung (Clay)	<0,004	Batulempung (Shale)

CATATAN PENTING

- Batuan sedimen sering berlapis.
- Mengandung fosil (umur relatif).
- Porositas & permeabilitas penting untuk eksplorasi minyak & gas.
- Dapat berubah menjadi batuan sedimen metamorf (mis. kuarsit, sabak).

GEOHERMAL CUTTING IDENTIFICATION GUIDE

LITHOLOGY • ALTERATION • MINERALIZATION • INDICATOR MINERALS

1. VOLCANIC ROCKS (PRIMARY LITHOLOGY)

ANDESITE  - Color : Gray – dark gray - Texture : Porphyritic – aphanitic - Hardness : Hard - Composition : Plagioclase, Pyroxene, Amphibole - Remarks : Most common host rock in geothermal system	BASALT  - Color : Dark gray – black - Texture : Aphanitic - Hardness : Very hard - Composition : Plagioclase, Pyroxene, Olivine - Remarks : Mafic volcanic rock, dense and compact	DACITE  - Color : Light gray - Texture : Porphyritic - Hardness : Hard - Composition : Plagioclase, Quartz, Biotite - Remarks : Intermediate to felsic volcanic rock	RHYOLITE  - Color : White – light gray - Texture : Aphanitic – porphyritic - Hardness : Hard - Composition : Quartz, Feldspar, Biotite - Remarks : Felsic volcanic rock, commonly welded
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2. PYROCLASTIC ROCKS

TUFF (ASH TUFF)  - Color : White – light gray - Texture : Fine grained, soft, often crumbly - Composition : Volcanic ash, glass, crystal fragments - Remarks : Common in geothermal area	LAPILLI TUFF  - Color : Gray – light gray - Texture : Lapilli size (2–64 mm) - Composition : Pumice, lithic fragments, crystal - Remarks : Pyroclastic deposit from explosive eruption	VOLCANIC BRECCIA  - Color : Heterogeneous - Texture : Angular fragments in a fine matrix - Composition : Various volcanic fragments - Remarks : Formed by eruption or tectonic brecciation	AGGLOMERATE  - Color : Brown – gray - Texture : Rounded fragments (>2 mm) - Composition : Volcanic clasts in tuffaceous matrix - Remarks : Pyroclastic sedimentary rock
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3. HYDROTHERMAL ALTERATION (SECONDARY)

FRESH (UNALTERED)  - Hard, compact, fresh appearance - No significant clay or secondary minerals - Sharp edges, crystalline texture	SILICIFIED  - Color : White – light gray - Very hard; glassy to sugary - Silica replacement - Common in upflow zones	ARGILLIC (CLAY ALTERATION)  - Color : White – yellow – brown - Soft, earthy, clayey - Composition : Kaolinite, Illite, Smectite - Typical in acid alteration zone	ADVANCED ARGILLIC  - Color : White – cream – brown - Silicified + clay minerals - Often with alunite / pyrophyllite - High temperature alteration	PROPYLITIC (CHLORITE–EPIDOTE)  - Color : Green – dark green - Composition : Chlorite, Epidote, Calcite - Common in peripheral zones	CARBONATE ALTERATION  - Color : White – cream - Effervescent with HCl - Composition : Calcite, Dolomite - Common in neutral–alkaline alteration
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4. COMMON VEINS & MINERALS

QUARTZ  - Color : White – clear - Hard, non-effervescent - Common in veins & fractures - Indicator of hydrothermal activity	CALCITE  - Color : White – cream - Effervescent with HCl - Common in veins & fracture fillings - Indicator of lower temperature fluid	PYRITE  - Color : Brass yellow - Metallic luster - Common in hydrothermal system - Indicator of reducing environment	HEMATITE / LIMONITE  - Color : Red – brown - Earthy to hard - Oxidation product - Common near surface or in oxidized zones
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5. ALTERATION MINERAL INDICATORS

CHLORITE  - Color : Green - Soft, flaky - Common in propylitic alteration - Indicator of moderate temperature	EPIDOTE  - Color : Pistachio green - Hard, granular - Common in propylitic alteration - Indicator of higher temperature	SMECTITE / CLAY  - Color : White – beige - Very soft, plastic when wet - Common in argillic alteration - Indicator of low temperature or surface alteration
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6. GRAIN SIZE REFERENCE

BOULDER > 256 mm	
COBBLE 64 – 256 mm	
PEBBLE 4 – 64 mm	
GRANULE 2 – 4 mm	
SAND 0.0625 – 2 mm	
SILT / CLAY < 0.0625 mm	

NOTES FOR MUD LOGGING

- Collect cutting every 1–3 meters.
- Wash with clean water and dry before description.
- Describe : Lithology, Color, Grain size, Texture, Hardness, Alteration, Minerals, Veins, Fracture.
- Take photo (macro) and keep sample properly.
- Correlate with drilling data (ROP, Gas, Mud Loss, Temperature, Pore Pressure).

HYDROTHERMAL ALTERATION ZONATION (GENERAL MODEL)



ABBREVIATION & SYMBOL

Qtz = Quartz	Sm = Smectite	 Vein
Cal = Calcite	Alu = Alunite	 Fracture
Py = Pyrite	Hmt = Hematite	 Fault
Chl = Chlorite	Pyro = Pyrophyllite	 Flow banding
Epi = Epidote	G = Grain	
Ill = Illite	Frac = Fracture	

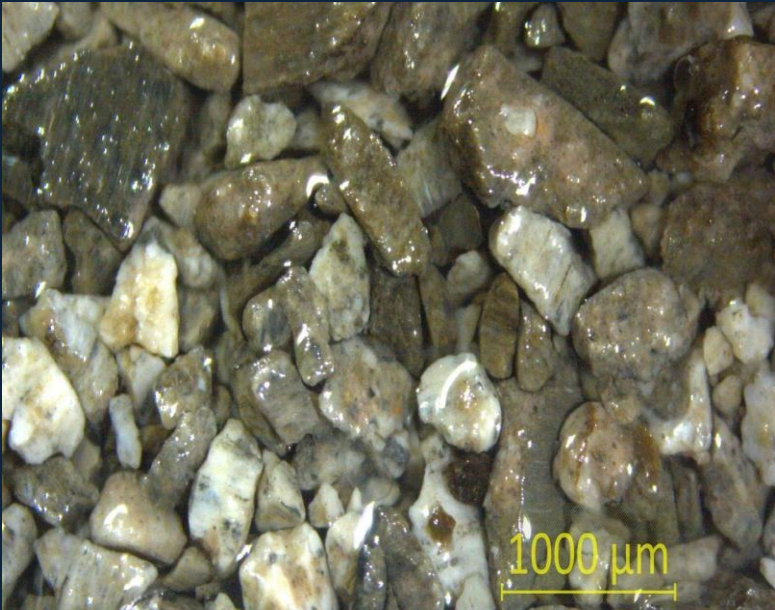
COLOR GUIDE (GENERAL)

COLOR	POSSIBLE INDICATION
White – Light Gray	: Silicification, Quartz, Alunite
Yellow – Brown	: Clay alteration, Oxidation
Green	: Chlorite, Epidote (Propylitic)
Red – Orange	: Hematite, Oxidation
Dark Gray – Black	: Basaltic, Mafic rocks
Cream – Beige	: Carbonate, Clay minerals

Cutting Analysis

Perbandingan cutting geothermal vs oil & gas

Parameter	 Geothermal	 Oil & Gas
Target utama	Reservoir panas bumi, fracture & feed zone	Reservoir hidrokarbon
Batuan dominan	Sering vulkanik, volcanoclastic, intrusive dan batuan teralterasi	Sering sedimentary rock
Contoh cutting	Andesite, Basalt, Dacite, Tuff, Volcanic Breccia	Sandstone, Shale, Limestone, Dolomite
Alteration	Sangat penting	Umumnya bukan parameter utama
Mineral sekunder	Chlorite, Epidote, Calcite, Quartz, Clay, Pyrite, Silica	Calcite, Dolomite, Pyrite, Quartz, Clay dan mineral diagenetik
Fracture	Sangat penting , dapat berhubungan dengan feed zone	Penting terutama pada fractured reservoir
Hydrocarbon show	Bukan target utama	Sangat penting
Oil stain	Umumnya tidak menjadi target	Dicatat
Fluorescence	Umumnya tidak menjadi target	Sangat penting untuk oil show
Cut test	Umumnya tidak dilakukan sebagai analisis utama	Umum untuk evaluasi oil show
Porosity	Penting, tetapi fracture permeability sering lebih dominan	Sangat penting untuk reservoir
Temperature indication	Sangat penting	Bukan fokus utama cutting
Mud loss	Sangat penting sebagai indikasi permeability/fracture	Penting, tetapi interpretasinya tergantung operasi
Gas	Dapat dimonitor, termasuk gas geothermal tertentu H2S, CO2	C1–C5 menjadi parameter utama mud logging, H2S, CO2
Tujuan deskripsi	Lithology + alteration + mineral + fracture/feed zone	Lithology + reservoir quality + hydrocarbon sho



Cutting Analysis - Drilling Oil & Gas

- Secara Umum diskripsi cutting/sample (wash sample) dibawah microscope meliputi ;
 - a. Rock Type
 - b. Classification
 - c. Color
 - d. Hardness
 - e. Grain Size
 - f. Grain Shape
 - g. Sorting
 - h. Minerals
 - i. Matrix pore filling
 - j. Cement
 - k. Visual Structure
 - l. Visual Porosity Estimation
 - m. Oil Show
 - n. Fossil
 - o. Other

Cuttings Lithological Description and Interpretation ❖ Cuttings Samples Description Equipment.

❖ Cuttings Samples Lithological Description.

❖ Cuttings Samples Lithological Interpretations.

❖ Rate of Penetration (ROP).

❖ Accessory Minerals.

❖ Visual Porosity.

❖ Well Geological Correlation.

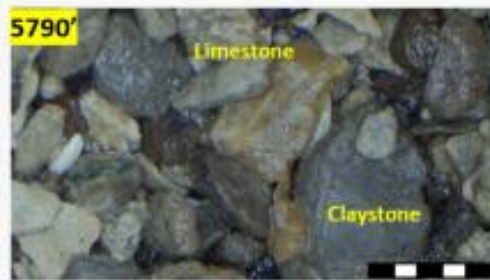
Examine the cuttings using abinocular microscope, UV box and laboratory tests to determine their composition, properties, hydrocarbon content, and visual porosity. The information is used to update computerized Masterlog.



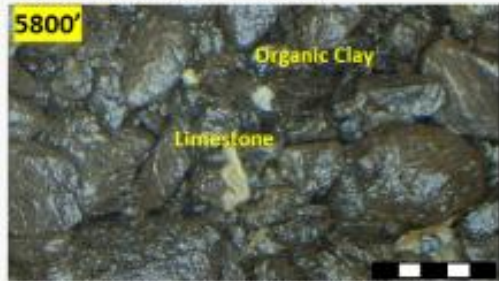
Cutting Analysis Drilling Oil & Gas



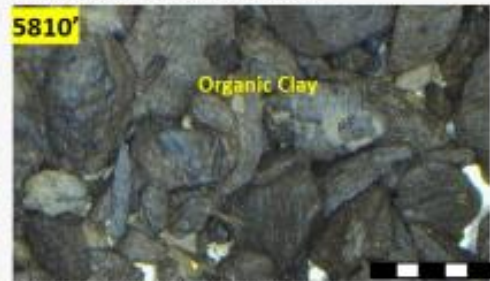
80% Claystone, 20% Limestone



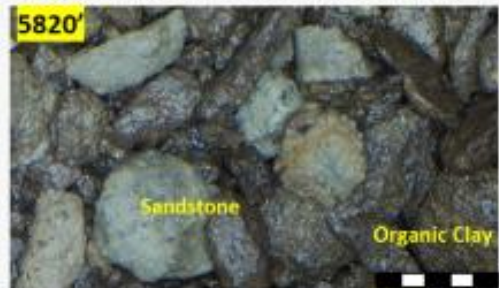
80% Claystone, 20% Limestone



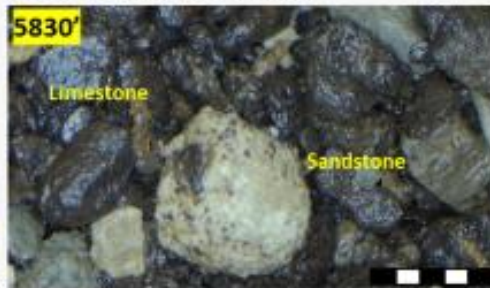
90% Claystone (Organic Clay), 10% Limestone



100% Claystone (Organic Clay)



80% Claystone (Organic Clay), 20% Sandstone



80% Claystone (Organic Clay), 20% Sandstone



90% Claystone, 10% Sandstone



100% Claystone



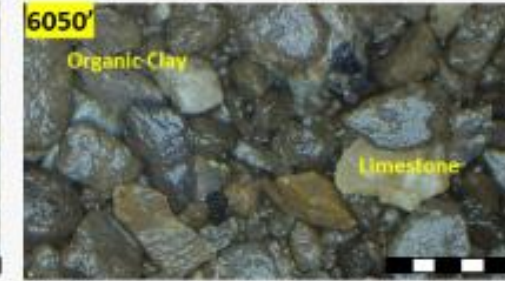
90% Claystone (Organic Clay), 10% Limestone



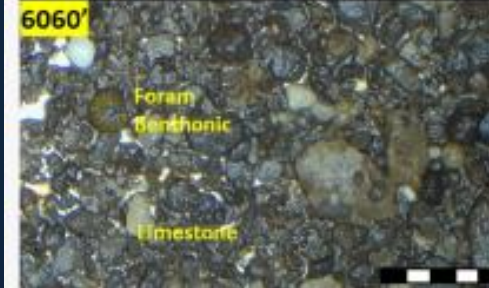
80% Claystone (Organic Clay), 20% Limestone



80% Claystone (Organic Clay), 20% Limestone



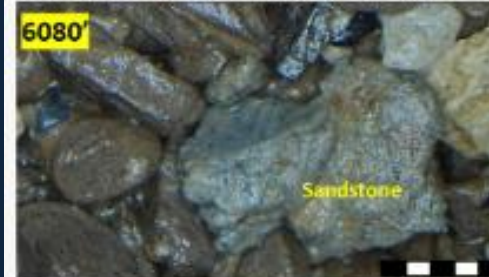
90% Claystone (Organic Clay), 10% Limestone



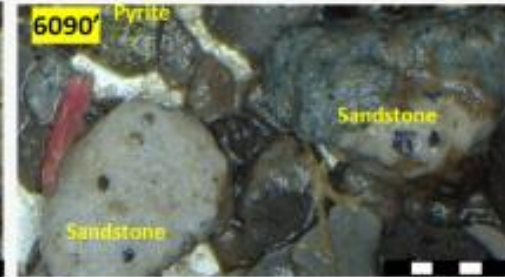
80% Claystone, 15% Sandstone, 5% Limestone



70% Claystone, 30% Sandstone



50% Claystone, 50% Sandstone



40% Claystone, 60% Limestone

HYDROCARBON ANALYSIS

Analysis Under Microscope

1. Oil In Mud

Amount, Color, Odor, Fluorescent -----

- Shale Shaker and Unwashed sample.

Pengamatan untuk minyak dimulai dari sample yang belum dicuci. Proses pengambilan sample dan pengamatan dengan mikroskop harus melalui evaluasi minyak yang sistematik.

2. Odor (Bau)

Biasanya batuan yang mengandung hidrokarbon mempunyai bau yang spesifik. Kekuatan baunya tergantung dari jenis dan kadar kuantitas kandungan hidrokarbon didalam batuan. Bau wangi biasanya berasal dari minyak parafine dan naftatik, sedangkan bau busuk berasal dari minyak aromatic. None, Poor (slight), Fair, Good

3. Oil Staining

Cuttings yang diduga ada oil stainnya (oily luster, yellow, or brownish discoloration) dipisahkan dan diletakkan pada spot plate yang bersih (hanya satu cutting per spot). Percentage under Microscope

Kenampakkan Oil Stain

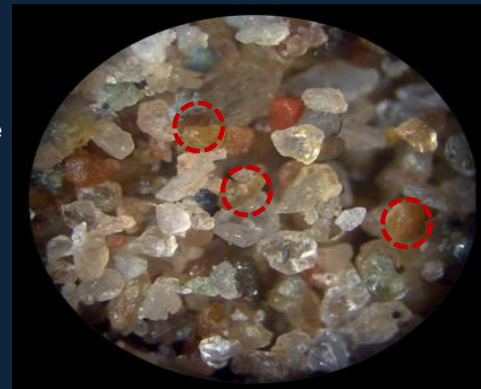
- ✓ No visible oil stain (tidak ada kenampakkan)
- ✓ Pin point oil stain (1-2 grain)
- ✓ Spotty oil stain (berbintik – bintik)
- ✓ Streaky oil stain (bergaris)
- ✓ Patchy oil stain (berlapis)
- ✓ Even oil stain (rata/seragam)
- ✓ Bleeding ois tain (penuh dengan oil)

Colour of stain

- ✓ No Visible oil stain
- ✓ Light brown oil stain
- ✓ Medium brown oil stain
- ✓ Dark brown oil stain
- ✓ Black asphaltic residu (dead oil) 90 – 100 % Exelent

Presentage oil stain

- | | |
|-------------|-------|
| ✓ 50 – 90 % | Good |
| ✓ 30 – 50 % | Fair |
| ✓ 10 - 30 % | Poor |
| ✓ < 10 % | Trace |



Analysis Under Ultra Violet Box (UV Box)

Untuk mengetahui kenampakan warna fluorescence

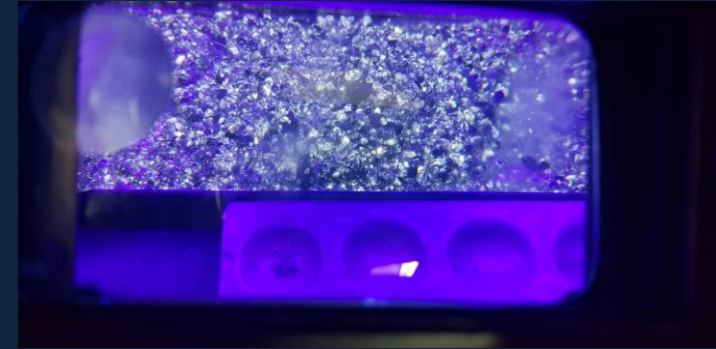
4. Fluorescence Color

Brown
Orange brown
Brownish gold
Orange
Gold
Yellowish orange
Yellow
Whitish yellow
Greenish yellow
Yellowish white
Bluish white



Heavy Oil

Light Oil



5. Cut Fluorescence Color

Only use Chlorothene

Bluish white
Yellowish white
Greenish yellow
Whitish yellow
Yellow
Yellowish orange
Gold
Orange
Brown
Brownish gold



Light Oil

Heavy Oil

6. Character (sifat) Cut Fluorescence Color

Flash Cut : hidrokarbon sangat cepat keluar dari kepingan batuan dan langsung larut dalam cairan solvent.

Cloudy Cut : Jejak hidrokarbon yang keluar memberikan kesan berkabut di dalam solvent.

Streaming Cut : Hidrokarbon yang keluar berjalan lambat di dalam solvent.

Crush Cut : hidrokarbon keluar dari kepingan batuan setelah di hancurkan.

Bleeding Cut : Seperti gelembung ketika diberikan larutan solvent

Kecepatan Reaksi Solvent

Fast (Flash Cut)

Moderate (Cloudy Cut)

Slow (Streamer Cut)

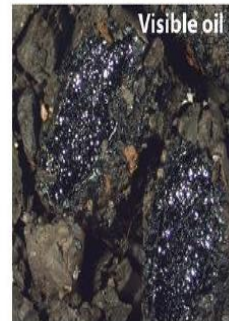
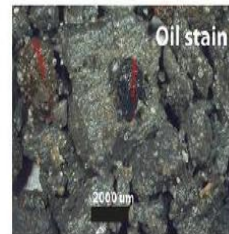
Very Slow (Crust cut)



Cutting Analysis - Drilling Oil & Gas

❖ Oil Shows Evaluation Steps:

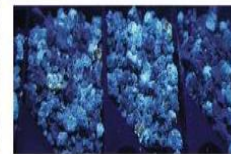
- **Step 1: OIL STAIN; Under Microscope:** (rr, tr, occ, mn, com, abdnt, pred) + (lt brn, brn, dk brn) **o stn or n vis o stn.**
- **Step 2: FLUORESCENCE DISTRIBUTION; Under UV Box:** (rr, tr, occ) spt/ (mn, com) patchy/ (abdnt, pred) even + (dl, fnt, bri) + (yel, yelshwh, wh, mlkywh....) **fluor.**
- **Step 3: CUT FLUORESCENCE; By adding solvent (Chloroform)** (rr, tr, occ, mn, com, abdnt, pred) + (slo/mod / fst) + (strmg/blmg) + (dl, fnt, bri) + (yel, wh, mlkywh) **ct fluor.**
- **Step 4: RESIDUAL OIL; Dissolving sample by adding solvent to sample on a filter paper:** (rr, tr, occ, mn, com, abdnt, pred) + (lt brn, brn, dk brn) + **resd o or n vis resd o.**
- **Oil Odor:** The first indication of the presence of hydrocarbon the odor (smell) may present in the cutting; you can use general term such as slight, mod, or strong oil odor.



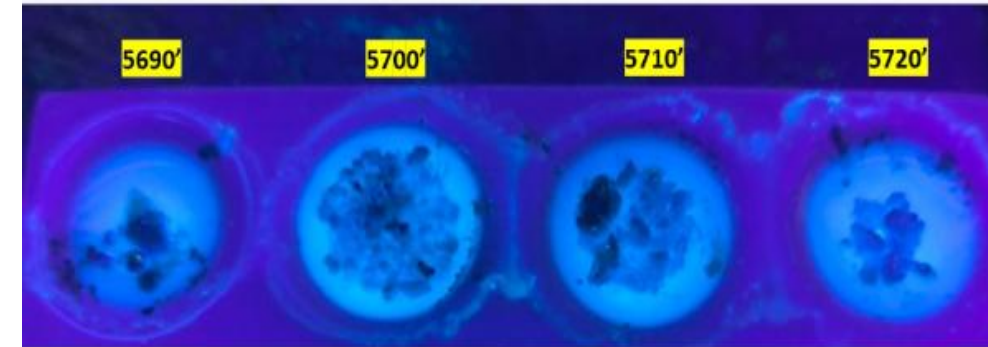
Oil cut



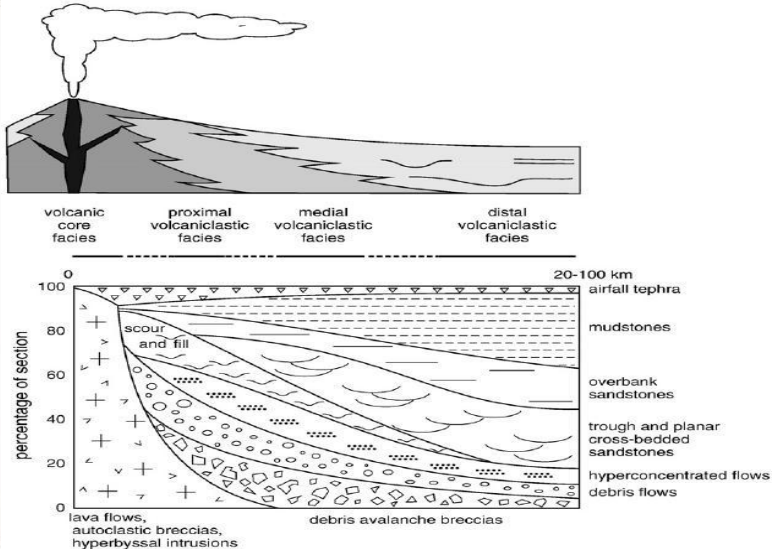
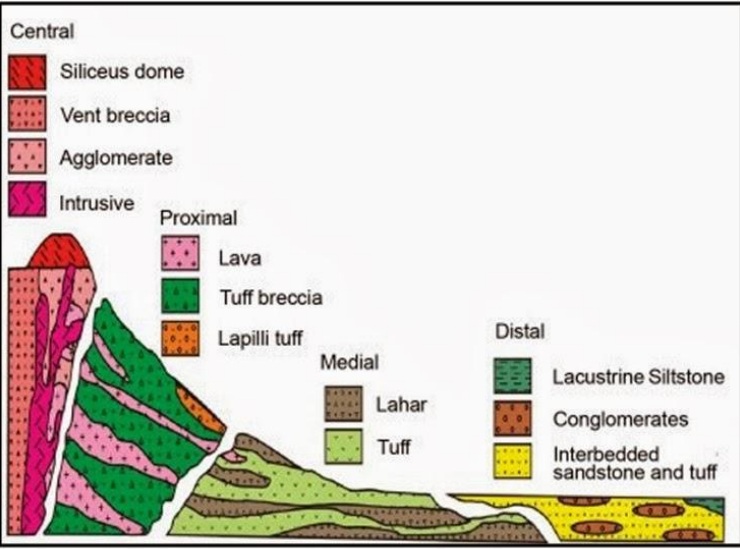
Fluorescence



Depth		Show Type	Show Quality	Descriptions
Top	Base			
5680	5690	Oil Trace	Poor	Bright yellow direct fluorescence, very slow streaming - thin milky white cut fluorescence, faint white residual ring.
5690	5700	Oil Fair	Fair	Bright yellow direct fluorescence, slow streaming - moderate thick milky white cut fluorescence, thin white residual ring.
5700	5710	Oil Fair	Fair	Bright yellow direct fluorescence, slow streaming - moderate thick milky white cut fluorescence, thin white residual ring.
5710	5720	Oil Poor	Poor	Bright yellow direct fluorescence, slow streaming - thin milky white cut fluorescence, faint white residual ring.



Model interpretasi volcanostratigraphy

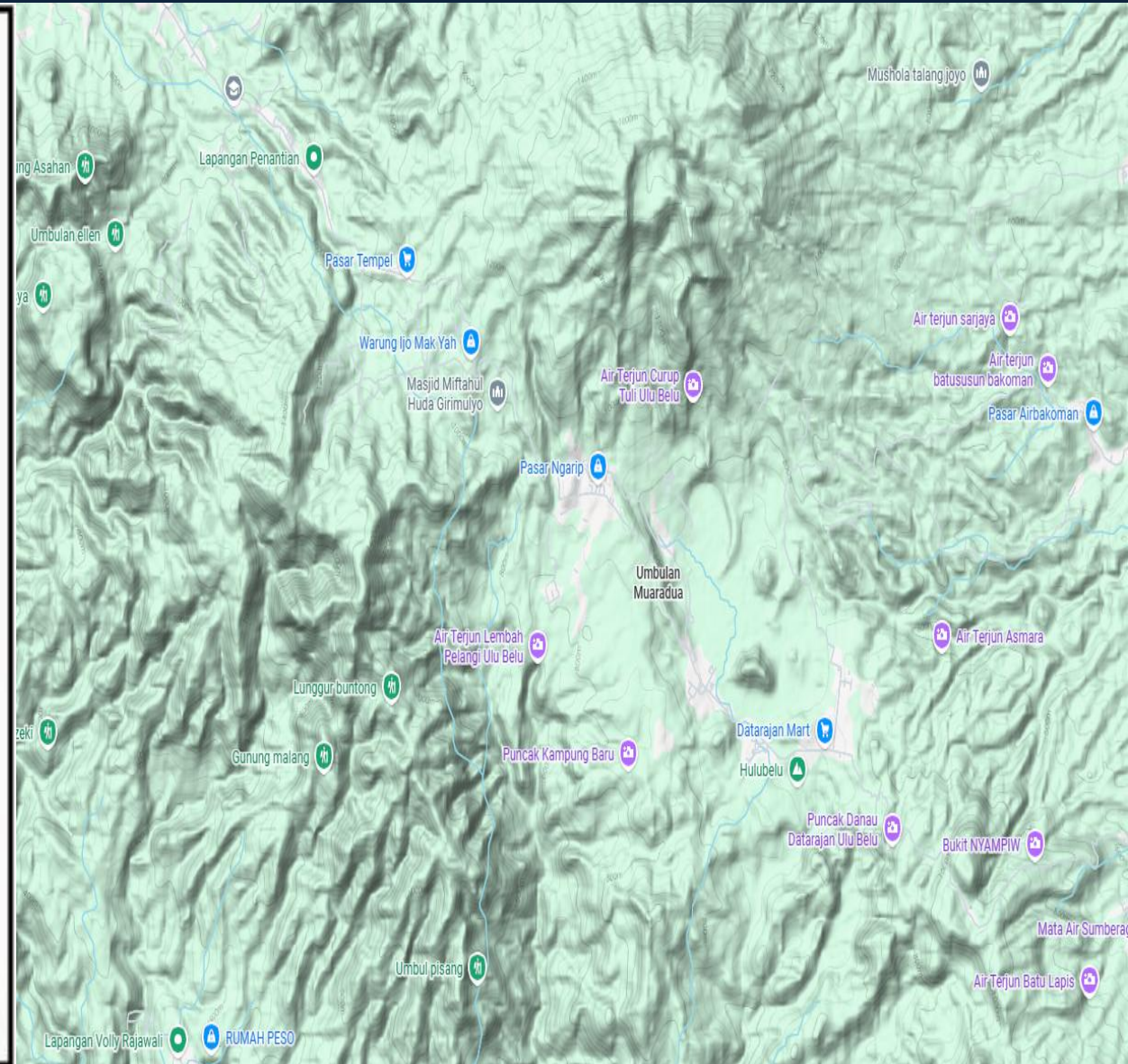
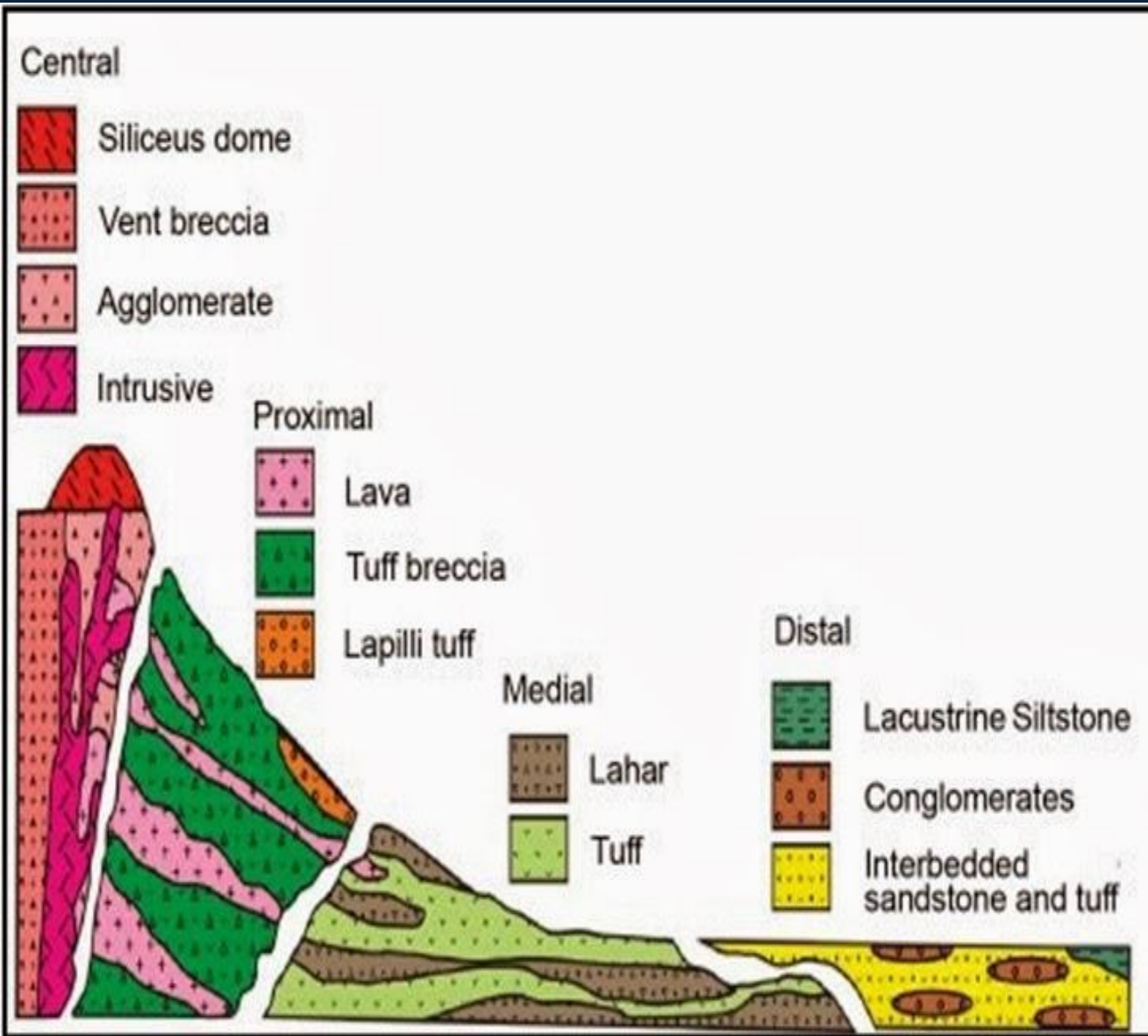


Zona/Fasies	Posisi	Litologi yang diperkirakan	Karakter utama	Signifikansi geothermal
Central / Vent	Pusat vulkanik	Intrusive, vent breccia, agglomerate, silicic dome	Sangat kasar, brecciated, banyak struktur	Potensi fracture & jalur fluida
Proximal	Lereng dekat pusat	Lava, lava breccia, tuff breccia, lapilli tuff	Lava tebal, coarse pyroclastic	Fracture pada lava/breccia dapat menjadi reservoir
Medial	Lereng menengah	Tuff, lapilli tuff, lahar, volcanoclastic	Material semakin halus, reworked	Umumnya transisi reservoir-cap
Distal	Kaki gunung/cekungan	Fine tuff, sandstone, conglomerate, siltstone/lacustrine	Sedimentasi/reworking dominan	Dapat membentuk aquifer atau lapisan penutup

Fasies / Zona	Litologi umum	Potensi masalah drilling	Indikator di rig/mud logging
Central / Vent	Intrusive, vent breccia, agglomerate, silicified rock	Severe lost circulation , fractured zone, hard formation, bit wear, vibration, stuck pipe	Total/partial losses, ROP fluktuatif, torque tinggi, cutting angular, quartz/pyrite/alteration kuat
Proximal	Lava, tuff breccia, volcanic breccia, lapilli tuff	Loss circulation, differential/stuck tendency, unstable breccia, bit bouncing, pack-off	Losses naik, cavings, torque/drag meningkat, ROP berubah tajam
Medial	Tuff, lapilli tuff, lahar, volcanoclastic	Hole instability , swelling/dispersive clay jika teralterasi, washout, pack-off, tight hole	Cutting lunak, clay tinggi, hole cleaning buruk, torque drag meningkat
Distal	Fine tuff, sandstone, conglomerate, siltstone/lacustrine	Sloughing, washout, sand production, shallow loss/gain, poor hole cleaning	Banyak fines, cavings, mud properties berubah, ROP tinggi namun hole quality buruk



Medan
Lihat topografi dan elevasi



Thank you

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